

Ganesh Gupta

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EXPERIENCE

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Took full ownership of **25+ production financial-transaction services** for **1M+ users** at **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**: defined problems, made **architecture decisions**, and shipped without waiting on specs.
- Led **architecture and code-review** sessions across production services, **mentoring engineers** on system design, scalability, and reliability tradeoffs.
- Standardized **retry, timeout, and failure-propagation** contracts across 25+ services, cutting production escalations **60%** in one quarter via **root cause analysis** and post-incident reviews.
- Redesigned a nightly **financial settlement aggregation** pipeline from **8 minutes to 90 seconds** via **indexed views** and batch partitioning, eliminating row-level lock contention at scale.
- Partnered directly with enterprise clients to translate billing and transaction-processing requirements into **technical designs** and custom integrations.

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Implemented **OAuth**-based auth and **RBAC** for secure, multi-tenant access to confidential pipeline data; participated in **design reviews** and **code reviews** on distributed service architecture.
- Engineered a **distributed executor** orchestrating pipeline-node lifecycle via typed **REST APIs**, backed by an **S3** artifact store at **sub-200ms p99 latency**, operated end-to-end from design through deployment.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Rebuilt a synchronous service into an **async, autoscaled** architecture on **GCP Cloud Run**, cutting **p99 latency from 420ms to 140ms** across **50K+ daily requests**.
- Triaged **distributed-system incidents** end-to-end on an **on-call rotation**, driving operational improvements with automated **CI/CD rollback**.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/Distributed-KV-Store

- Built a replicated key-value store in **Go** implementing **Raft consensus**, **linearizable reads**, and **log compaction**, tolerating node failures without data loss.

VPC Control Plane Simulator

github.com/ganeshhgupta/vpc-control-plane-sim

- Built an AWS-style **VPC** networking simulator with **CIDR** validation, **security-group** evaluation, and configuration **drift detection**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built a self-hosted observability platform tracing requests, **latency**, and cost through a **FastAPI** collector and **PostgreSQL** store.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Java, Python, Go, SQL, C++

Distributed Systems: Architecture & Design Reviews, Fault-Tolerant Placement, Raft Consensus, OAuth/RBAC

Cloud & Infra: AWS (EC2, RDS, SQS), GCP, Docker, CI/CD

Leadership: Mentoring, Code Review, Root Cause Analysis, Cross-Team Technical Partnership

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer